Temperatures at four interior points of a square plate

Step-by-step COMSOL Multiphysics 4.3a guide for students

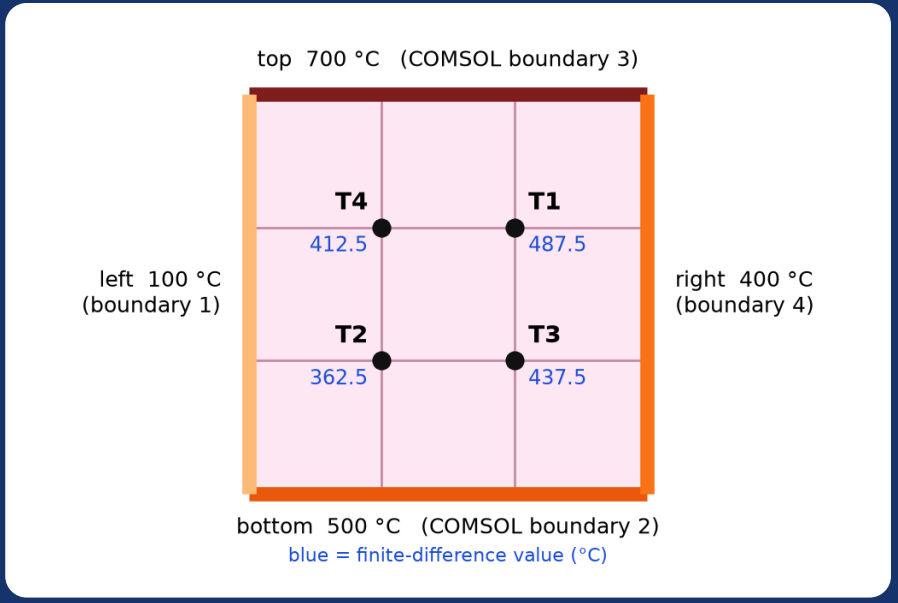
Problem: 2D steady heat conduction, 3 x 3 grid, Finite Difference Method (Introduction to CFD, Problem 3)

Tool: COMSOL Multiphysics 4.3a, Heat Transfer in Solids, 2D stationary study with a mesh-size sweep

Result (finite difference, 4 unknowns): T1 = 487.5, T2 = 362.5, T3 = 437.5, T4 = 412.5 degC

COMSOL, converged mesh: 490.36, 359.64, 438.07, 411.93 degC = exact series

Includes: hand solution (Gauss, Gauss-Seidel, heat balance), 16 annotated COMSOL steps, grid study



Model file: plate_fdm_comsol.mph (open it in COMSOL to compare with your own model)

The problem and the finite difference setup

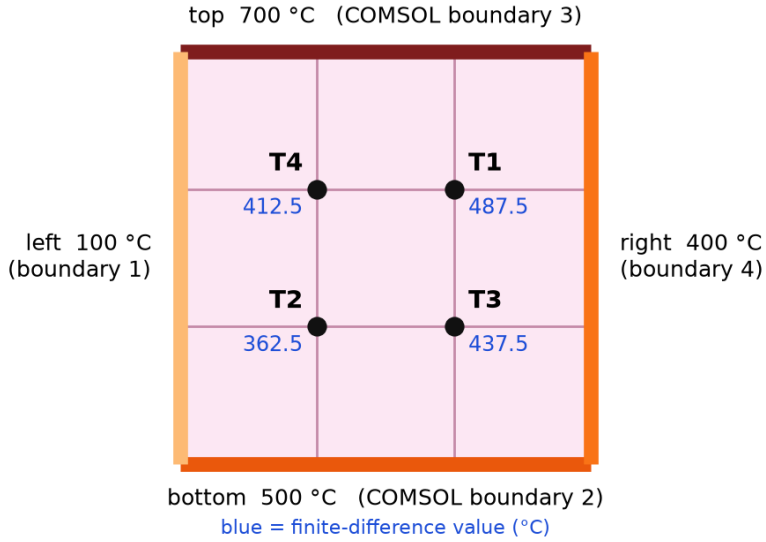
Calculate the temperatures at points 1, 2, 3 and 4 of the square plate using the Finite Difference Method. The square is divided into 3 x 3 equal squares (Delta x = Delta y). The edges are held at 700 C (top), 100 C (left), 400 C (right) and 500 C (bottom). The plate size and the conductivity are not given: they are not needed.

Edge	top	left	right	bottom
Temperature	700 C	100 C	400 C	500 C

Assumptions: steady state; 2D conduction (no variation through the thickness); no heat generation; constant conductivity k.

Tasks

1. Physics: governing equation and boundary conditions.
2. Grid: number the 4 x 4 nodes (i, j); known and unknown nodes.
3. Discretisation: five-point formula, four equations.
4. Solve: (a) Gaussian elimination, (b) Gauss-Seidel.
5. Post-processing: temperatures on the plate, heat in = heat out.
6. Grid study: finer grids, how close is the 4-node answer?



1. Physics: Laplace equation

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = 0$$

Boundary conditions: T = 700 (y = top), T = 500 (bottom), T = 100 (left), T = 400 (right).

2. Grid: 4 x 4 nodes, 3 x 3 squares

The 16 nodes (i, j), i, j = 1..4. The 12 nodes on the edges (T_i,1, T_i,4, T_1,j, T_4,j) are known. Only the 4 interior nodes are unknown: T1 = T(3,3), T2 = T(2,2), T3 = T(3,2), T4 = T(2,3). The 4 corner nodes are not needed: the stencil of an interior node never touches a corner.

3. Discretisation: five-point formula

Central differences, d2T/dx2 ~ (T(i+1,j) - 2 T(i,j) + T(i-1,j)) / dx2, and the same in y. With dx = dy:

$$T_{i,j} = \frac{1}{4}\left(T_{i+1,j} + T_{i-1,j} + T_{i,j+1} + T_{i,j-1}\right)$$

Each interior temperature is the average of its four neighbours. For the four nodes:

```
T1 = (700 + 400 + T4 + T3) / 4  ->  4 T1 - T3 - T4 = 1100
T2 = (100 + 500 + T4 + T3) / 4  ->  4 T2 - T3 - T4 =  600
T3 = (400 + 500 + T1 + T2) / 4  ->  4 T3 - T1 - T2 =  900
T4 = (700 + 100 + T1 + T2) / 4  ->  4 T4 - T1 - T2 =  800
```

Node 1 is the top-right node: its neighbours are the top edge (700), the right edge (400), node 4 (to its left) and node 3 (below it). The other three follow the same pattern.

Hand solution of the four equations

A. Gaussian elimination of [A | b]

```
start:        T1     T2     T3     T4   |    b
R1:            4      0     -1     -1   |  1100
R2:            0      4     -1     -1   |   600
R3:           -1     -1      4      0   |   900
R4:           -1     -1      0      4   |   800

column 1:  R3 <- R3 + R1/4,  R4 <- R4 + R1/4
R3:            0     -1   3.75  -0.25   |  1175
R4:            0     -1  -0.25   3.75   |  1075

column 2:  R3 <- R3 + R2/4,  R4 <- R4 + R2/4
R3:            0      0    3.5   -0.5   |  1325
R4:            0      0   -0.5    3.5   |  1225

column 3:  R4 <- R4 + R3/7
R4:            0      0      0 3.4286   |  1414.29
```

Back substitution:

```
T4 = 1414.29 / 3.4286              = 412.5
T3 = (1325 + 0.5 T4) / 3.5          = 437.5
T2 = (600 + T3 + T4) / 4            = 362.5
T1 = (1100 + T3 + T4) / 4           = 487.5
```

C. Heat balance (Task 5): heat through the boundary of the 4-node model

Edge	sum of k (T_edge - T_node)	direction
top (700)	(700-412.5) + (700-487.5) = 500.0	into the plate
bottom (500)	(500-362.5) + (500-437.5) = 200.0	into the plate
left (100)	(100-412.5) + (100-362.5) = -575.0	out of the plate
right (400)	(400-487.5) + (400-437.5) = -125.0	out of the plate

In: 500 + 200 = 700. Out: 575 + 125 = 700. In units of k x degC (Delta y / Delta x = 1). The balance is exact because adding the four node equations gives exactly this: the finite difference scheme conserves energy.

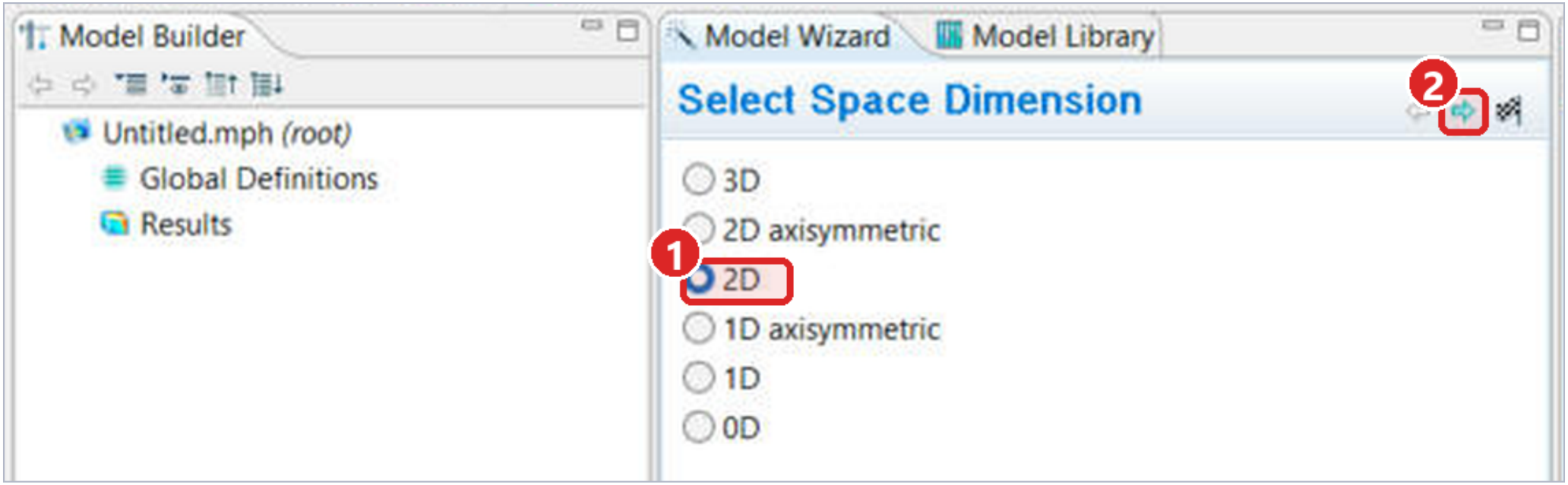
B. Gauss-Seidel (from T = 0; order T1,T4,T2,T3)

iter	T1	T2	T3	T4
0	0.00	0.00	0.00	0.00
1	275.00	217.19	348.05	268.75
2	429.20	327.41	414.15	361.60
3	468.94	353.31	430.56	399.09
4	482.41	359.87	435.57	408.93
5	486.13	361.77	436.97	411.50
6	487.12	362.30	437.35	412.22
7	487.39	362.44	437.46	412.42
8	487.47	362.48	437.49	412.48
12	487.50	362.50	437.50	412.50

Each new value uses the newest neighbours, e.g. iteration 1: T1 = (700 + 400 + 0 + 0)/4 = 275, then T4 = (700 + 100 + 275 + 0)/4 = 268.75, T2 = (100 + 500 + 268.75 + 0)/4 = 217.19, T3 = (400 + 500 + 275 + 217.19)/4 = 348.05. Stopping when no temperature changes by more than 0.001 degC takes 12 iterations (9 if we start from the mean edge temperature 425 degC). Gauss-Seidel and Gaussian elimination give the same answer.

Step 1 - Start a 2D model

Start COMSOL Multiphysics 4.3a. The Model Wizard opens (if not: File > New).

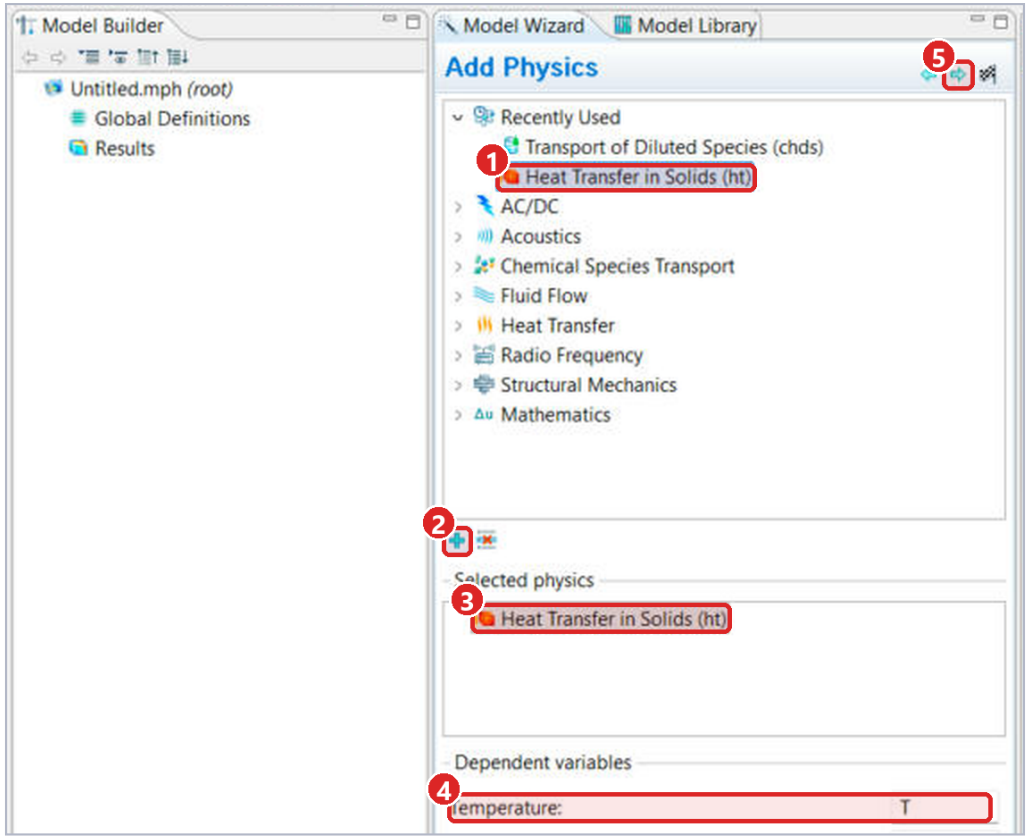


1. Select 2D. The plate is flat and the temperature does not change through its thickness (Assumption 2), so the temperature depends only on x and y.
2. Click the Next arrow.

Why?
A 2D model needs only the outline of the plate (a square). Heat conduction in a thin plate with insulated faces is exactly a 2D problem.

Step 2 - Add Heat Transfer in Solids

Steady heat conduction is described by the Heat Transfer in Solids interface (physics name ht).

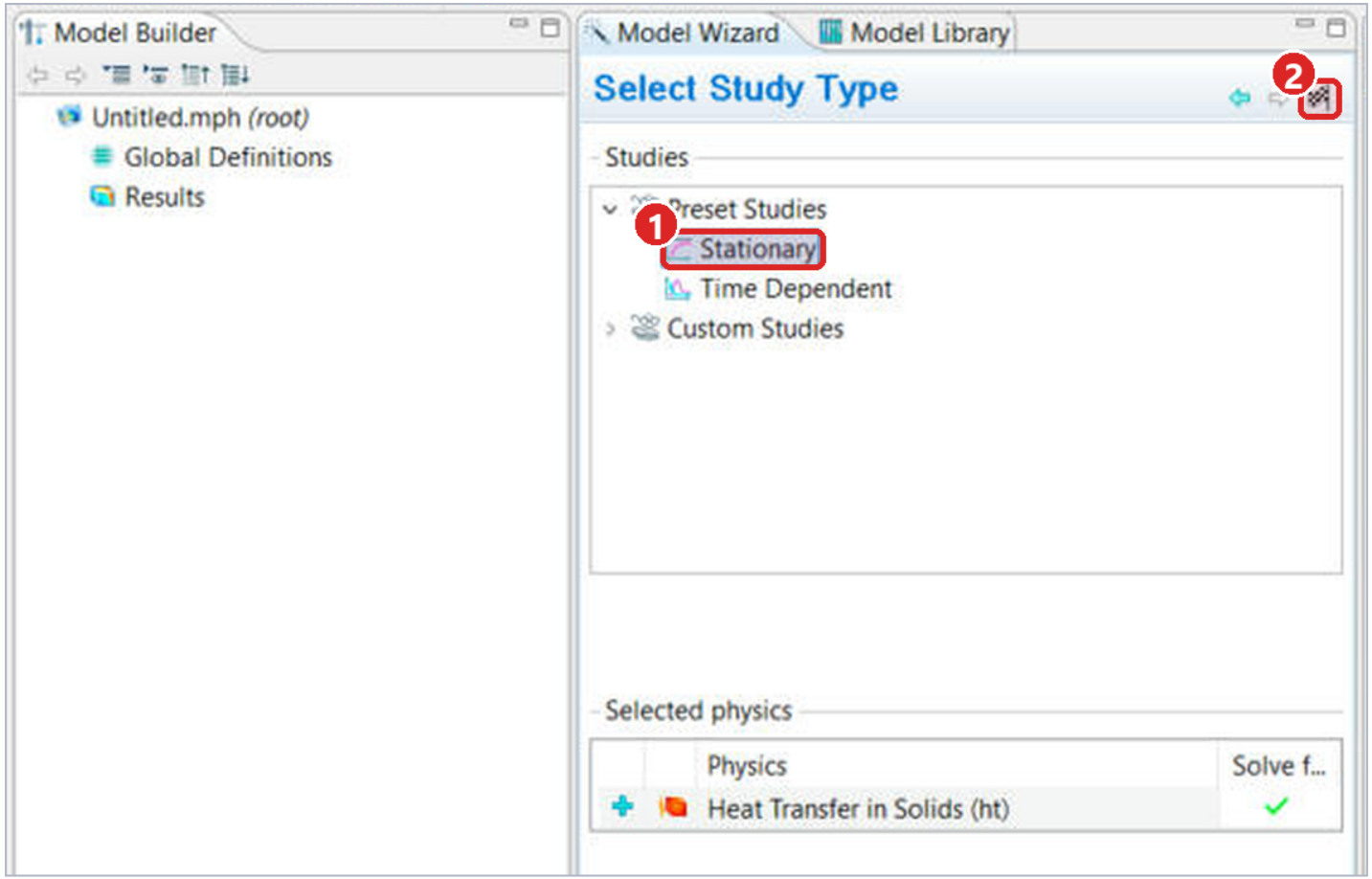


1. Select Heat Transfer in Solids (ht). It is in the folder Heat Transfer; on our screen it is also listed under Recently Used.
2. Click the blue + (Add Selected) button.
3. The interface appears under Selected physics.
4. The dependent variable is the temperature T (COMSOL stores it in kelvin).
5. Click Next.

Why?
COMSOL solves rho Cp dT/dt + div(-k grad T) = Q. In steady state (no time derivative), with no heat generation (Q = 0) and constant k, this becomes the Laplace equation d2T/dx2 + d2T/dy2 = 0 of Task 1.

Step 3 - Choose a stationary study

The temperatures no longer change with time (Assumption 1), so we need a steady-state study.

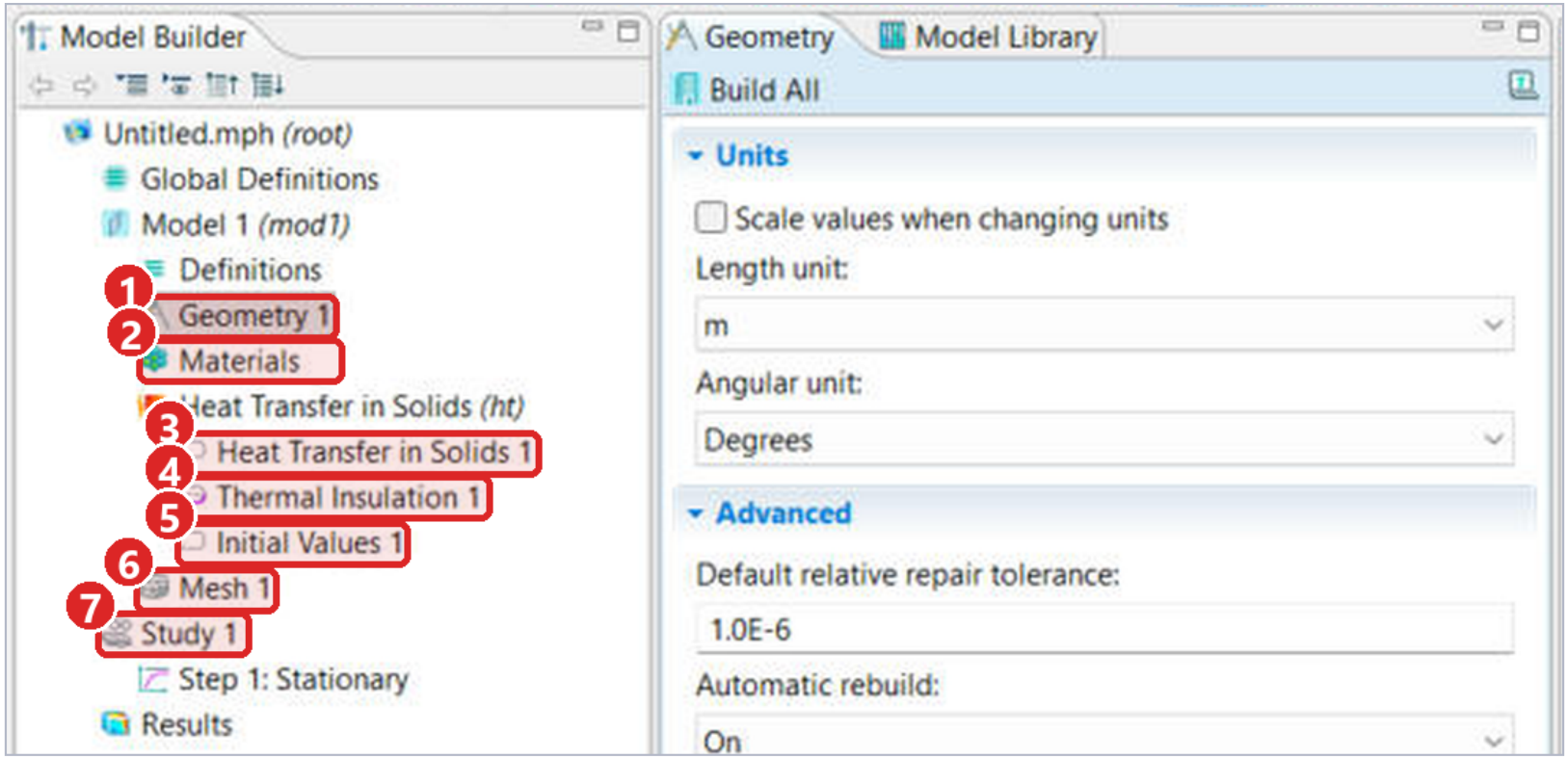


1. Select Stationary (under Preset Studies).
2. Click Finish (the chequered-flag icon).

Why?
A stationary study solves the whole discretised system A T = b in one go. This is what the finite difference method does by hand with Gauss elimination (Task 4a), except that COMSOL uses the finite element method.

Step 4 - Tour of the model tree

After Finish the Model Builder holds the model outline. Note what COMSOL has already created for us.

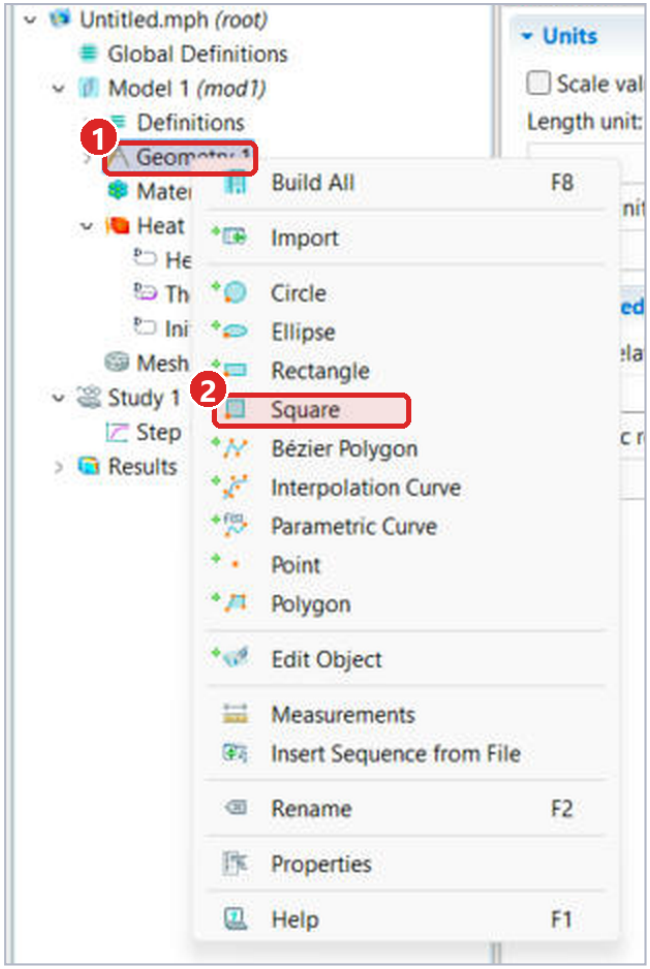


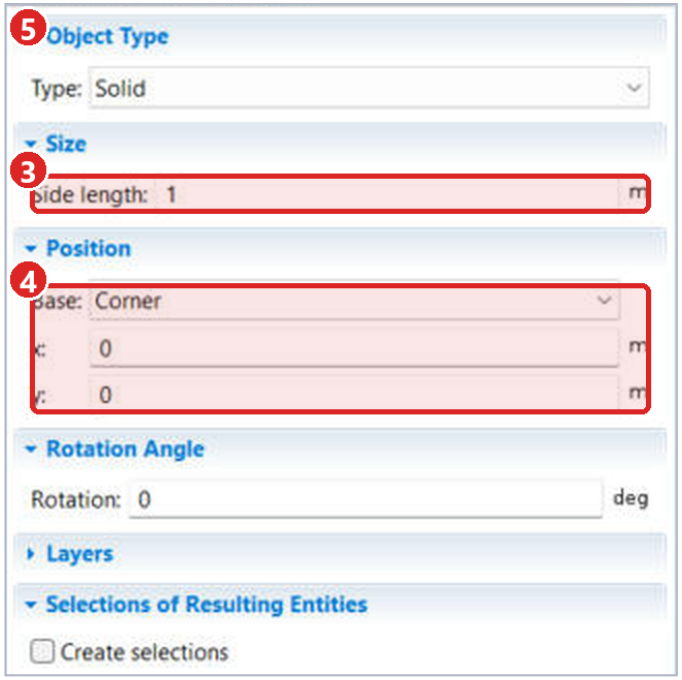
1. Geometry 1: the plate outline (length unit: metre).
2. Materials: we must supply the thermal conductivity k.
3. Heat Transfer in Solids 1: the equation. It takes k from the material.
4. Thermal Insulation 1: the default on every boundary. We will override it with Temperature conditions on all four edges.
5. Initial Values 1: not needed for a stationary study, but harmless.
6. Mesh 1: the grid (finite difference: Delta x = Delta y).
7. Study 1 with Step 1: Stationary.

Why?
The plate size and the conductivity are not given in the problem because they do not matter: the temperatures depend only on the four edge temperatures. We use a 1 m x 1 m plate with k = 1 W/(m K).

Step 5 - Build the geometry: a unit square

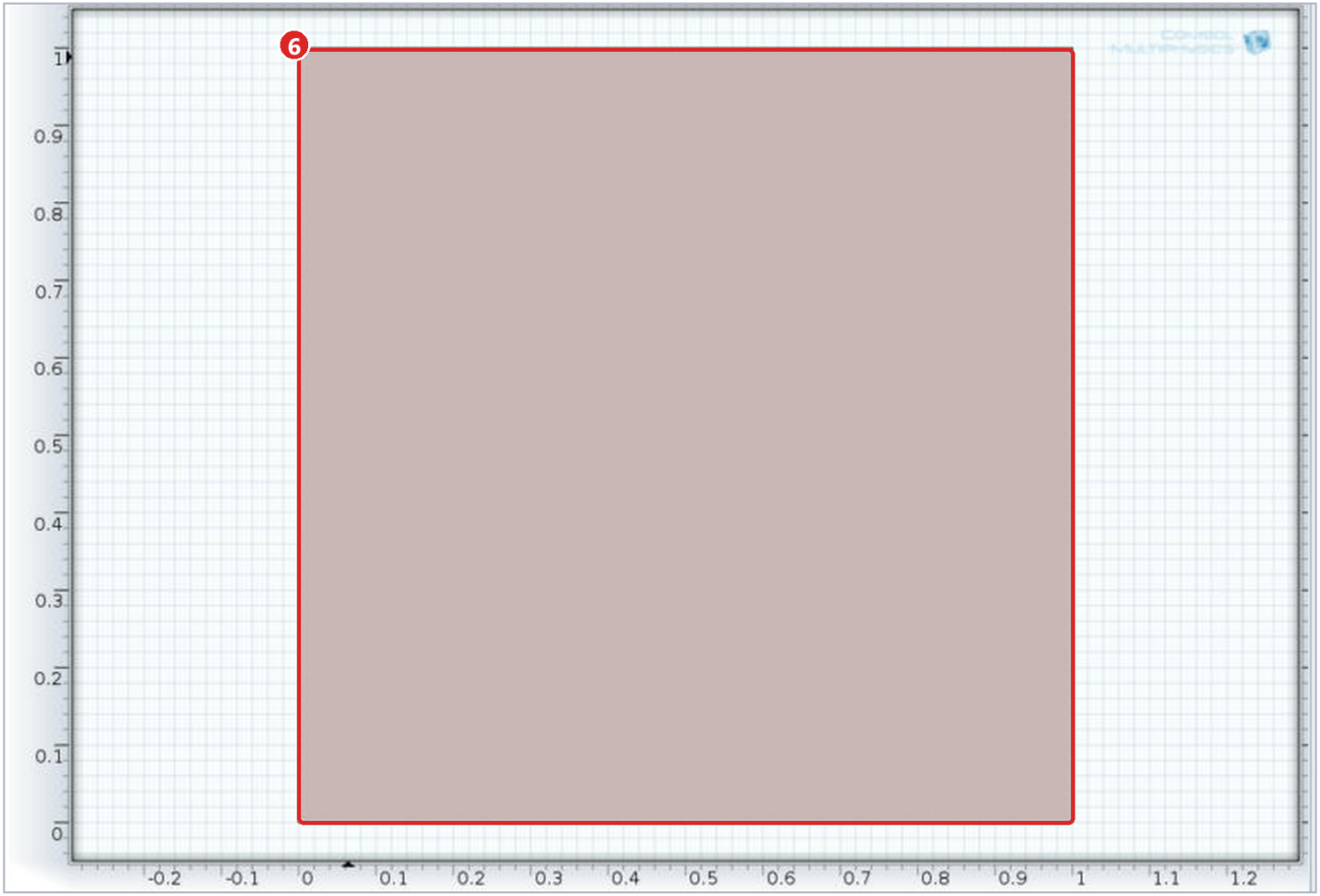
The plate is a square. Its size does not affect the temperatures, so we take 1 m x 1 m.





1. Right-click Geometry 1 (left-click it first if no menu appears).
2. Choose Square.
3. Side length = 1 m (the default).
4. Position: Base = Corner, x = 0, y = 0 (the defaults): the plate occupies 0 <= x <= 1 and 0 <= y <= 1.
5. Click Build Selected (or press F8).

Step 5 - Build the geometry: a unit square

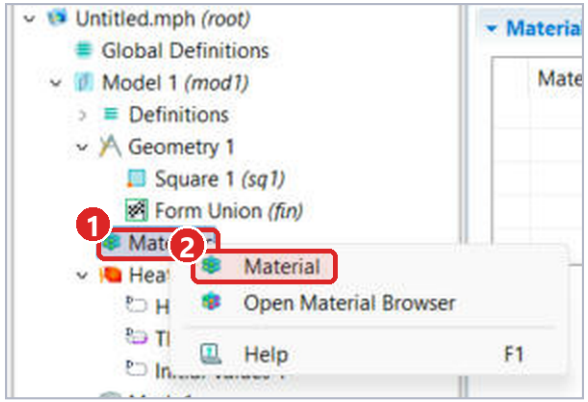


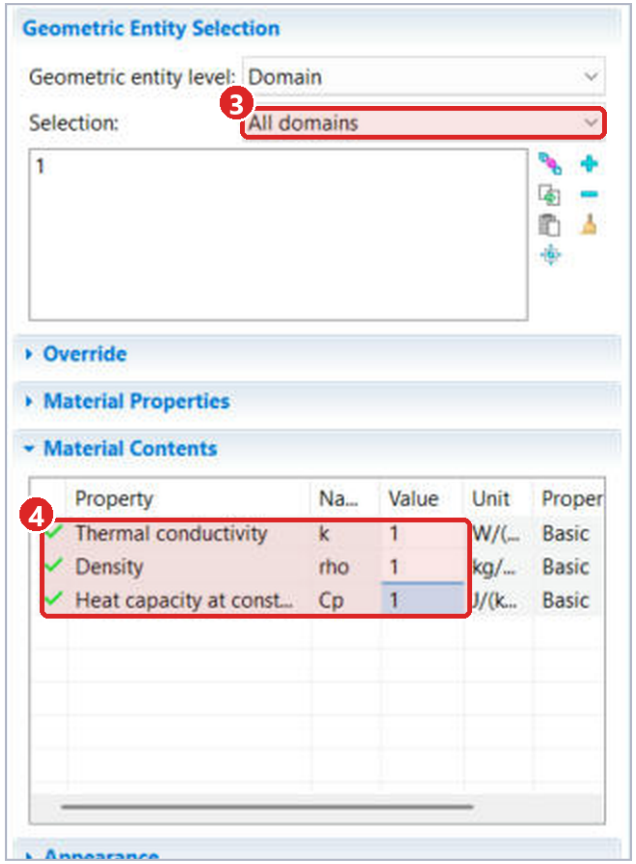
6 The square appears in the Graphics window. Its four edges are the four boundaries of the problem.

Why?
COMSOL builds the geometry in metres. The square has four boundaries and four corner points. The problem's Delta x = Delta y grid will come from the mesh size (Step 10).

Step 6 - Define the material (k = 1)

The heat equation needs a thermal conductivity. Because k is constant and there are no sources, it cancels out of the answer.



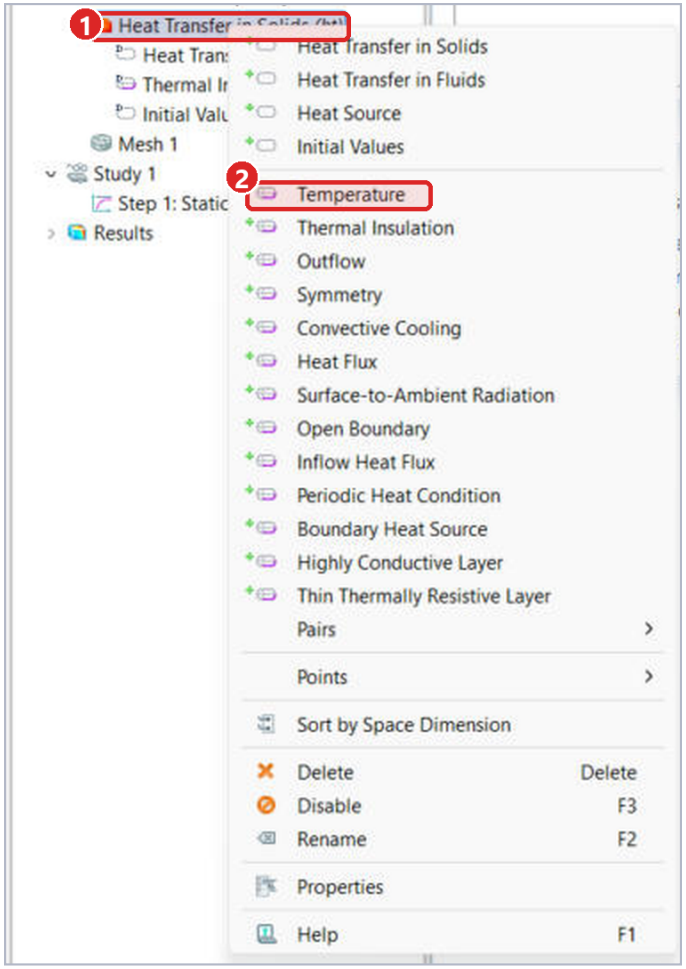


1. Right-click Materials (left-click it first).
2. Choose Material (a blank material).
3. Selection: All domains. The material covers the whole square.
4. In the Value column enter k = 1, rho = 1 and Cp = 1 (double-click a cell, type, press Enter). Each row then shows a green tick instead of a red icon.

Why?
For this steady, source-free problem only k enters the equation, and with constant k it can be any value without changing the temperatures. Density and heat capacity are not used by a stationary study, but COMSOL does not accept empty fields.

Step 7 - Temperature on the top edge: 700 C

Each edge is held at a known temperature. This is a Dirichlet (prescribed value) boundary condition.



1. Left-click Heat Transfer in Solids (ht), then right-click it.
2. Choose Temperature.

Step 7 - Temperature on the top edge: 700 C



3 The selection list must contain 3 (the top edge). Hover the cursor over the top edge until it turns red, click it, then press the + button (Add to Selection).

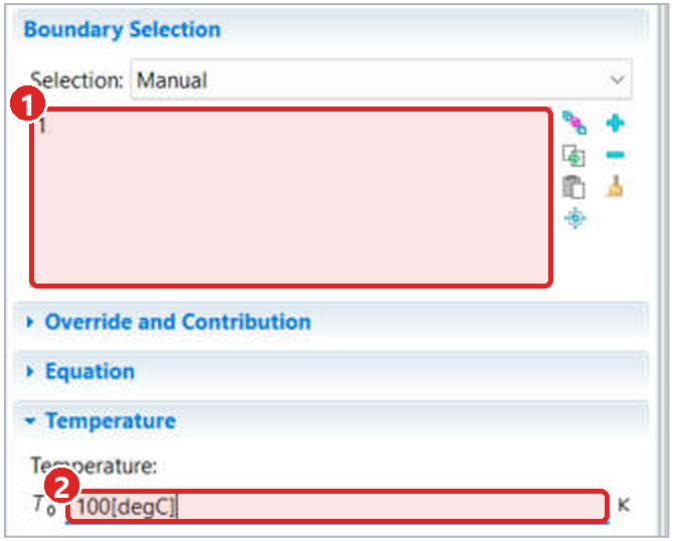
4 Temperature T0 = 700[degC]. Click in the field, press Ctrl+A to select the old text, type the new value and press Enter. COMSOL converts it to kelvin (973.15 K).

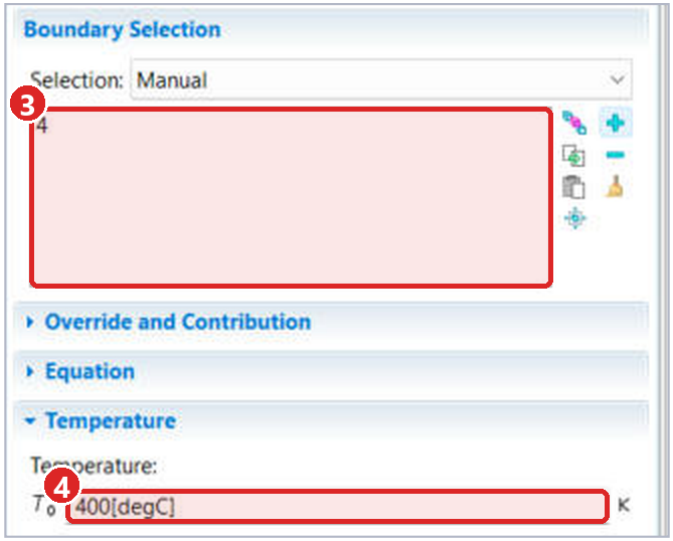
5 The selected edge is drawn in blue.

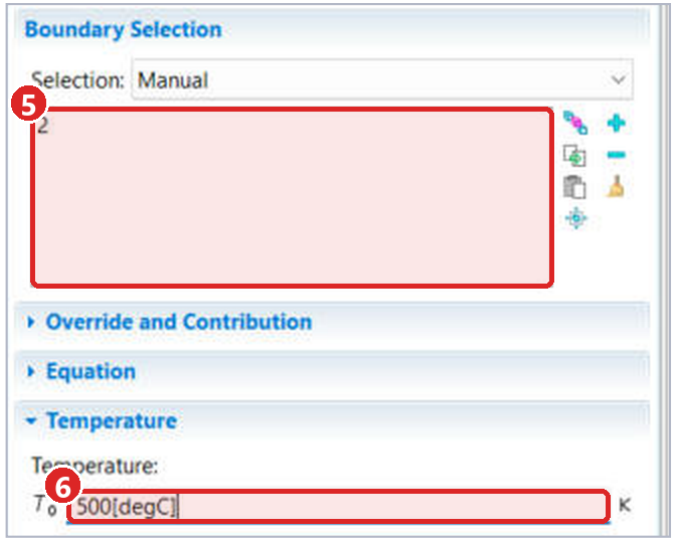
Why?
Edge numbers in this square: 1 = left, 2 = bottom, 3 = top, 4 = right. The temperature unit is written in square brackets: 700[degC]. Without it COMSOL would read 700 as 700 K.

Step 8 - Temperature on the left, right and bottom edges

Repeat Step 7 three more times: right-click ht > Temperature, select the edge, set the value.





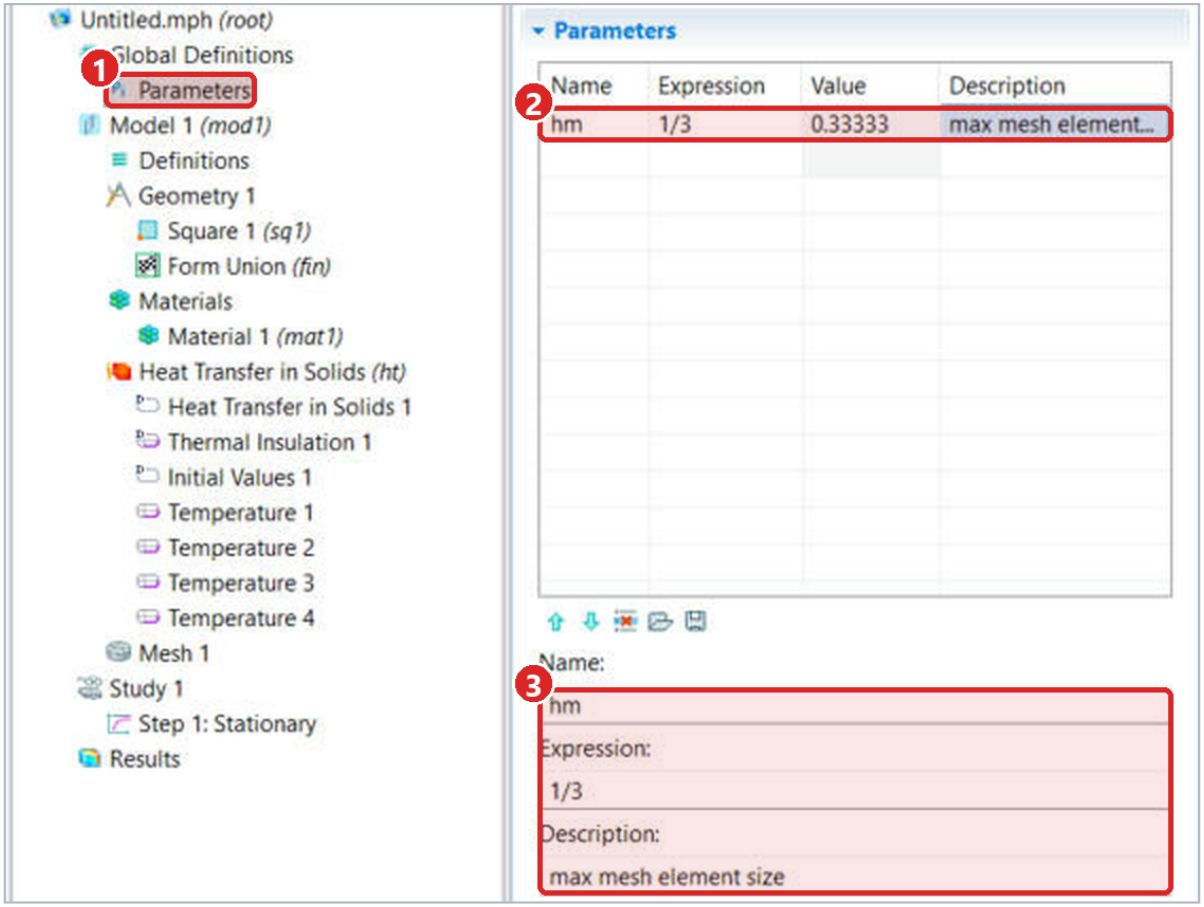


1. Temperature 2: selection list = 1 (the left edge).
2. T0 = 100[degC].
3. Temperature 3: selection list = 4 (the right edge).
4. T0 = 400[degC].
5. Temperature 4: selection list = 2 (the bottom edge).
6. T0 = 500[degC].

Why?
Check each Temperature node: the list must show exactly one edge and no edge may appear in two nodes. If a wrong edge is in the list, select it in the list and press the minus button. The four corner points belong to two edges each; the temperature there is not defined by the problem, and the interior temperatures do not need it.

Step 9 - Make the mesh size a parameter

We want to repeat the calculation on finer and finer meshes (Task 6). A parameter hm for the element size makes that easy.

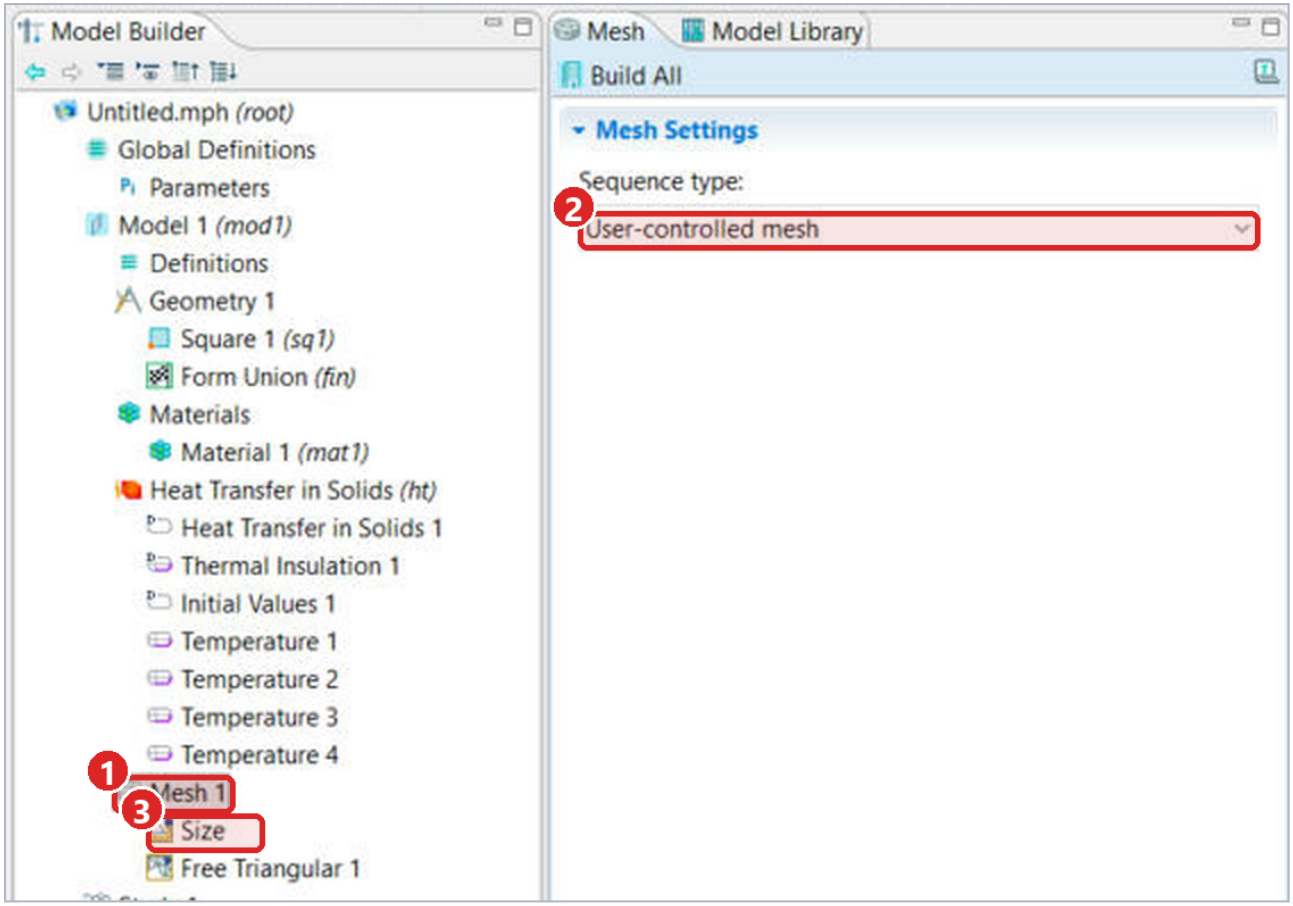


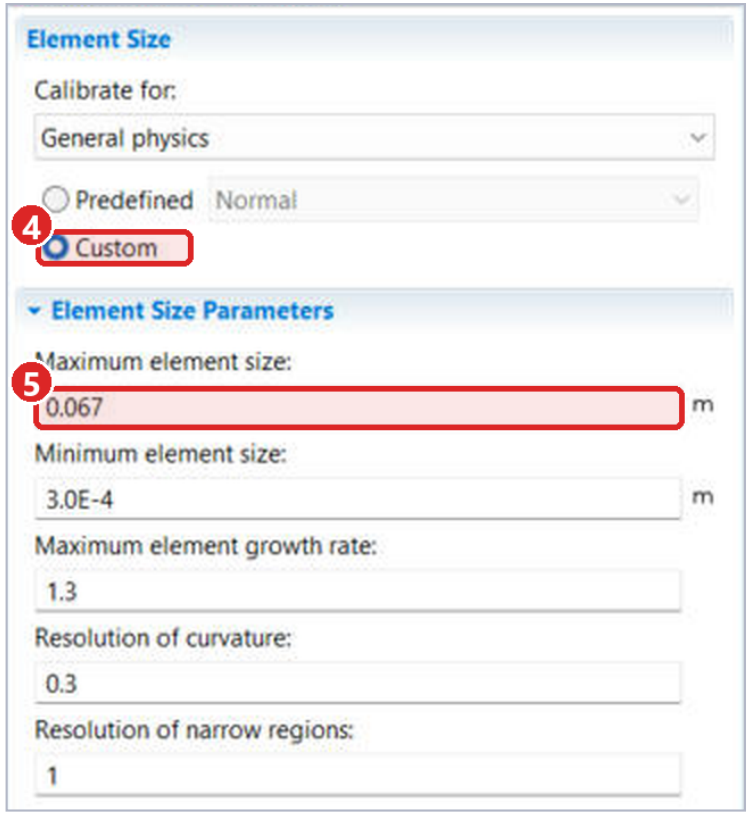
1. Right-click Global Definitions > Parameters (left-click Global Definitions first).
2. First row: Name hm, Expression 1/3, Description max mesh element size. Type the Name, press Tab, type the Expression, press Tab, type the Description, press Enter.
3. The fields below the table show the same row: hm = 1/3 (value 0.33333).

Why?
hm = 1/3 is the grid spacing Delta x = Delta y of the problem's 3 x 3 grid. Later we sweep hm over 1/3, 1/6, 1/12, 1/24, 1/48 and 1/96.

Step 10 - Create the mesh with maximum size hm

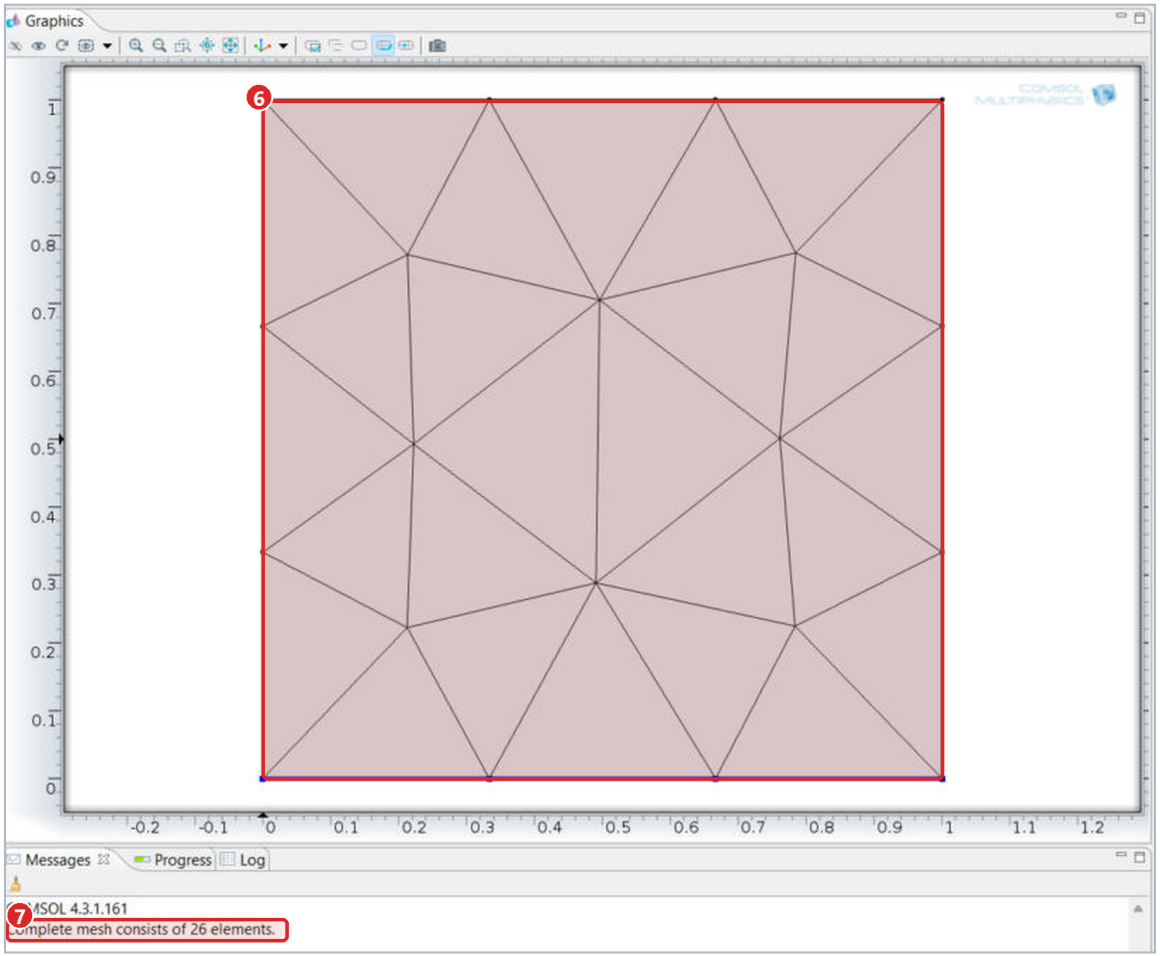
COMSOL meshes the square with triangles. We limit the element size with the parameter hm.





1. Click Mesh 1.
2. Sequence type: choose User-controlled mesh.
3. Two nodes appear: Size and Free Triangular 1. Click Size.
4. Select Custom (instead of Predefined).
5. Maximum element size = hm. Then click Build All (F8).

Step 10 - Create the mesh with maximum size hm



6 The mesh: 3 elements along every edge.

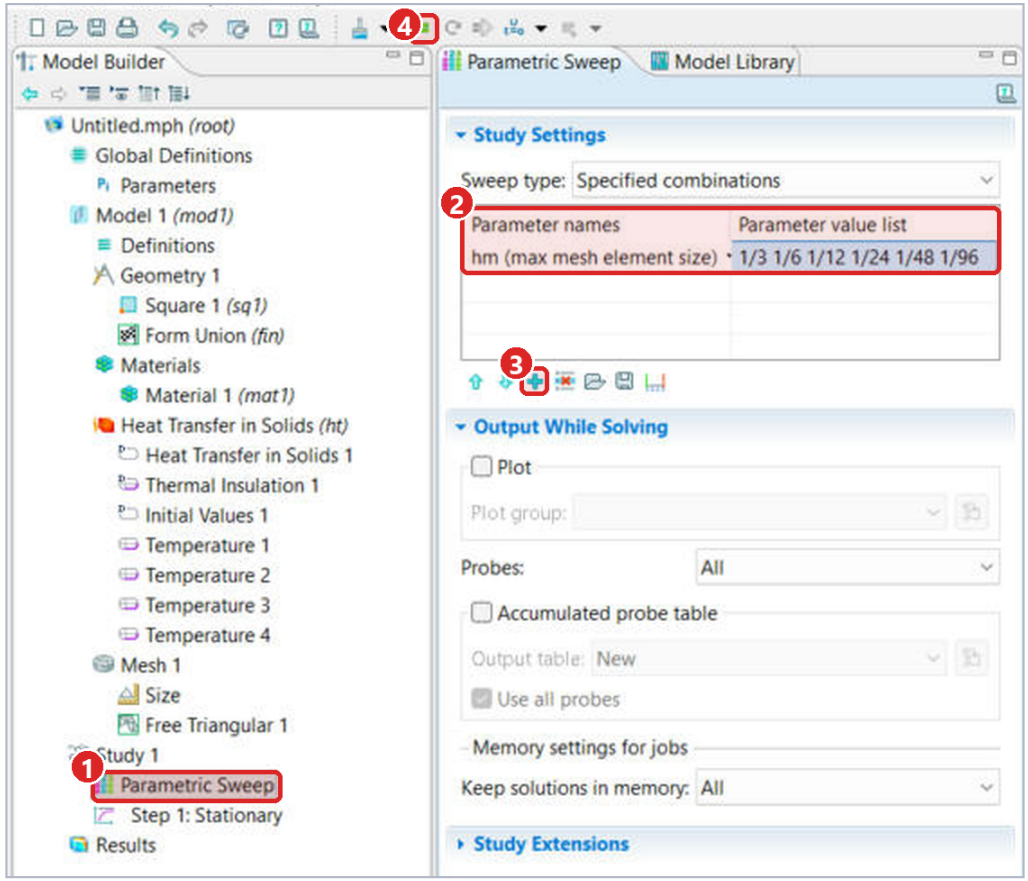
7 Messages: Complete mesh consists of 26 elements.

Why?

With hm = 1/3 each edge is divided into three, like the 3 x 3 grid of the problem. Unlike the finite difference grid, COMSOL uses triangles and quadratic shape functions, so its solution at the same spacing is already different. The parametric sweep in the next steps refines the mesh automatically.

Step 11 - Add a parametric sweep over hm

A parametric sweep solves the model once per value of hm, rebuilding the mesh each time.

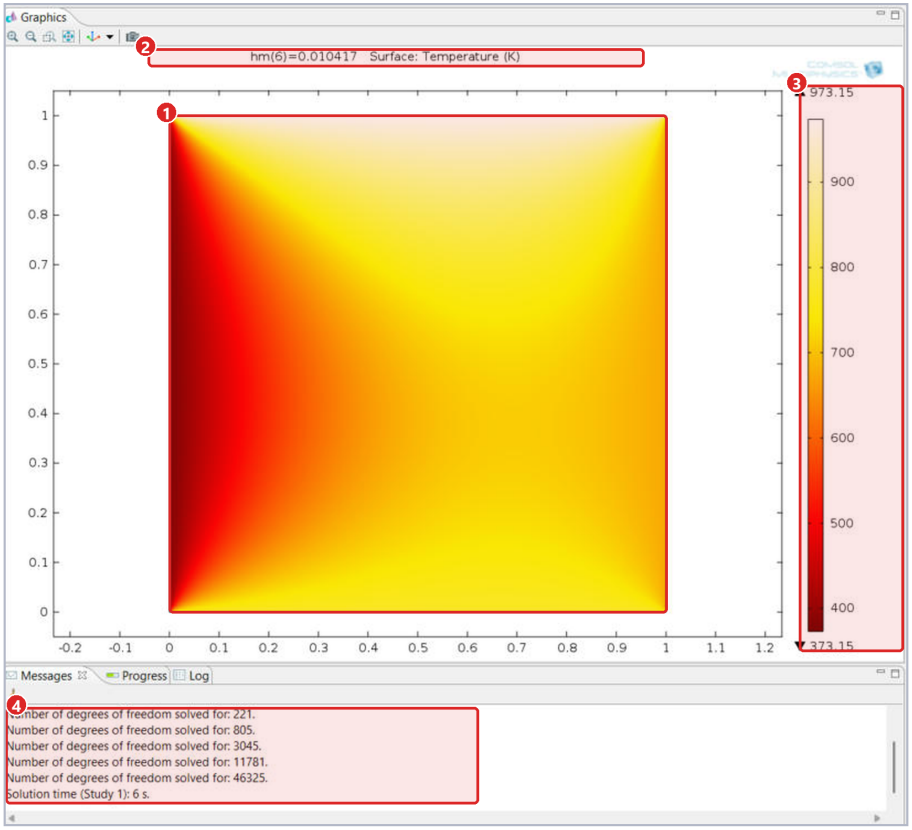


1. Right-click Study 1 > Parametric Sweep (left-click Study 1 first).
2. In the table: Parameter names = hm (choose it with the + button), Parameter value list = 1/3 1/6 1/12 1/24 1/48 1/96 (separate the values with spaces).
3. The + button adds a table row; COMSOL offers the parameters defined in Global Definitions.
4. Click Compute (the green = in the toolbar, or press F8).

Why?
Each value of hm halves the element size of the previous one. The finest mesh (hm = 1/96) has 46 325 degrees of freedom; the whole sweep takes about 6 s.

Step 12 - Compute and look at the temperature field

After Compute, COMSOL shows the temperature of the last parameter value (hm = 1/96) automatically.

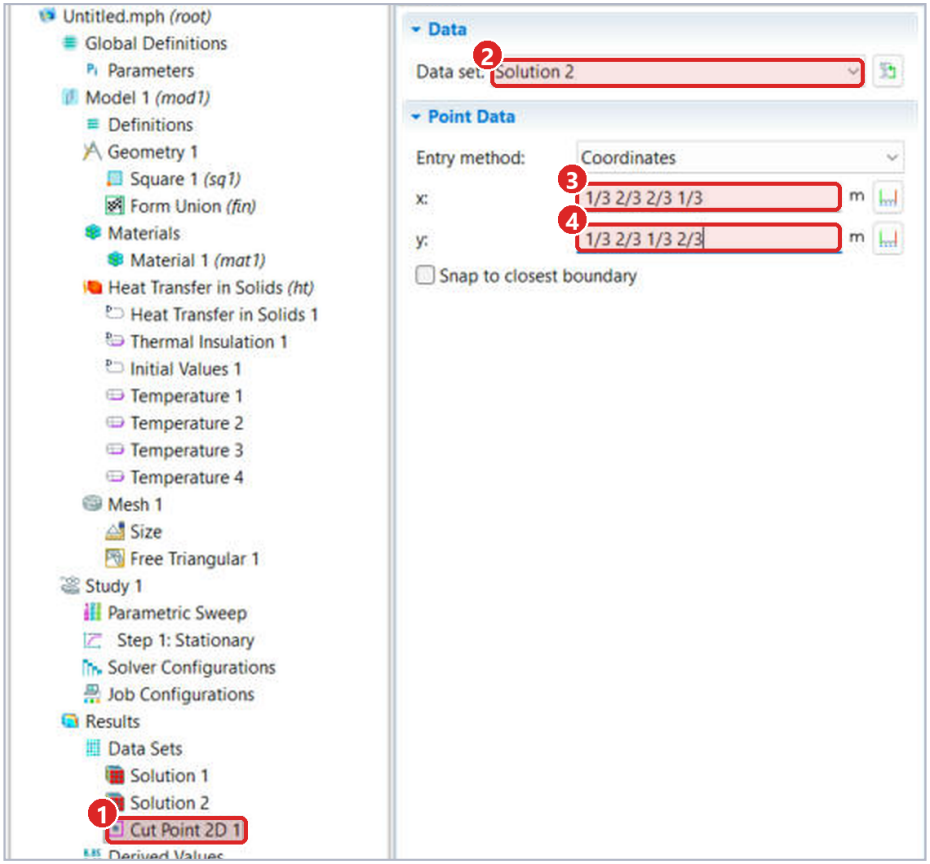


1. The temperature surface plot: hot top, cold left, and sharp corners.
2. The title states the parameter value: hm(6) = 0.010417 = 1/96, i.e. the 6th and finest mesh.
3. The colour legend is in kelvin: 373.15 K = 100 C (left edge) up to 973.15 K = 700 C (top edge).
4. Messages: degrees of freedom solved for 221, 805, 3045, 11781, 46325 for the finer meshes; Solution time (Study 1): 6 s.

Why?
The corners of the plate where two edges at different temperatures meet are singular (the temperature jumps there). The solution is smooth everywhere else, in particular at the four interior points that we need.

Step 13 - Read the temperature at the four points

The four interior points are at (1/3, 1/3), (2/3, 2/3), (2/3, 1/3) and (1/3, 2/3). We create a Cut Point 2D data set that contains them.

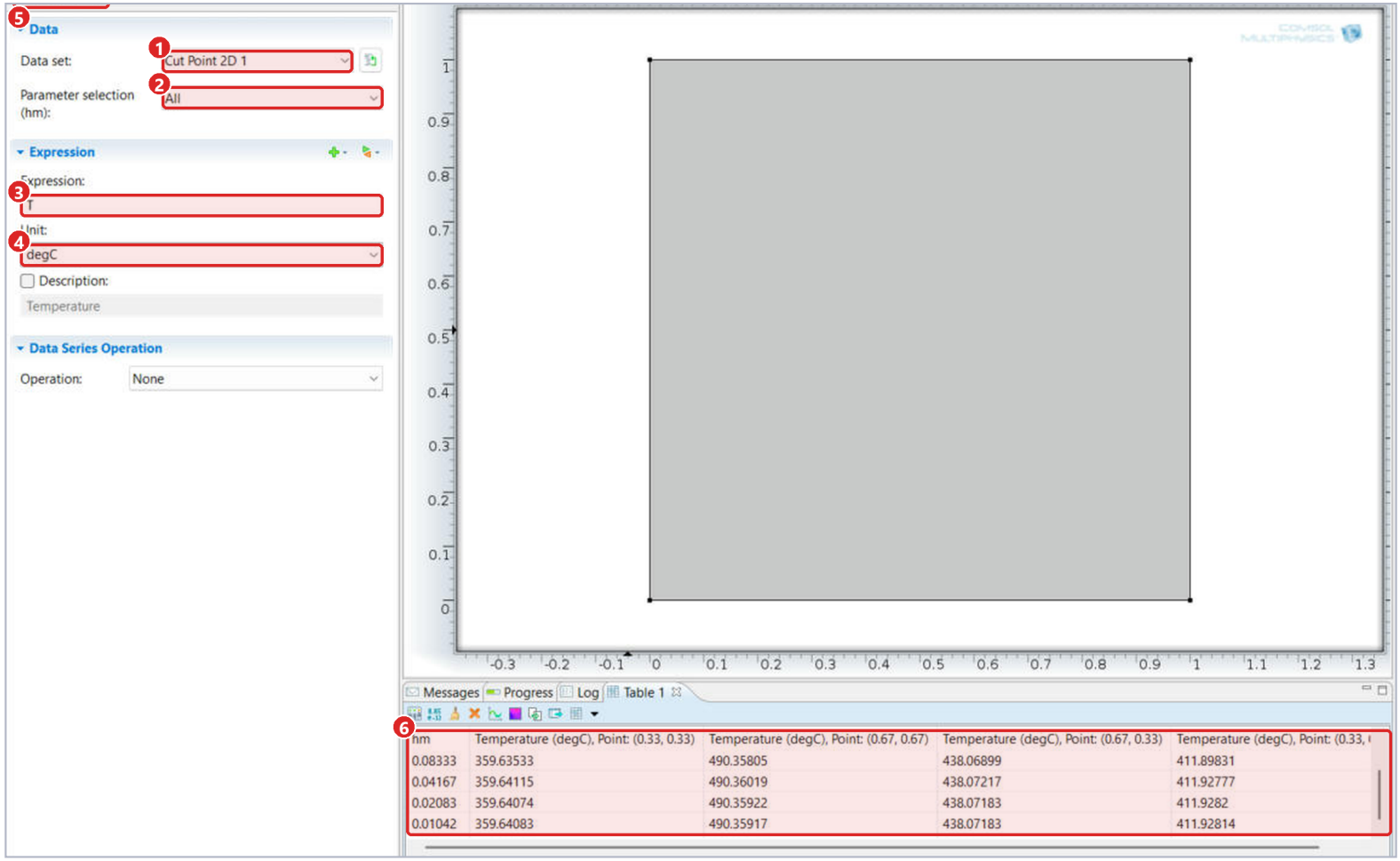


1. Right-click Results > Data Sets > Cut Point 2D. The node Cut Point 2D 1 appears.
2. Data set: Solution 2 (the parametric solutions, Solution 1 is only the first mesh).
3. x: 1/3 2/3 2/3 1/3
4. y: 1/3 2/3 1/3 2/3. Together: (1/3,1/3) = node 2, (2/3,2/3) = node 1, (2/3,1/3) = node 3, (1/3,2/3) = node 4 (the order of the columns later).

Why?
The x and y fields accept several values separated by spaces. Point number k uses the k-th x and the k-th y.

Step 14 - Evaluate the temperature (point evaluation)

A Point Evaluation puts the temperature at the four points into a table, one row per mesh.



1. Right-click Results > Derived Values > Point Evaluation. Data set: Cut Point 2D 1.
2. Parameter selection (hm): All, so every mesh gets a row.
3. Expression: T.
4. Unit: degC (type degC in the field; COMSOL otherwise shows kelvin).
5. Click Evaluate (the = at the top of the settings).
6. The table (Table 1) opens at the bottom; double-click its tab to enlarge it.

Step 14 - Evaluate the temperature (point evaluation)

7 hm	8 Temperature (degC), Point: (0.33, 0.33)	Temperature (degC), Point: (0.67, 0.67)	Temperature (degC), Point: (0.67, 0.33)	Temperature (degC), Point: (0.33, 0.67)
0.33333	358.81006	490.92252	438.45265	411.19642
0.16667	359.78585	490.38364	438.11403	411.74231
0.08333	359.63533	490.35805	438.06899	411.89831
0.04167	359.64115	490.36019	438.07217	411.92777
0.02083	359.64074	490.35922	438.07183	411.9282
0.01042	359.64083	490.35917	438.07183	411.92814

7 One row per value of hm: 1/3, 1/6, 1/12, 1/24, 1/48, 1/96.

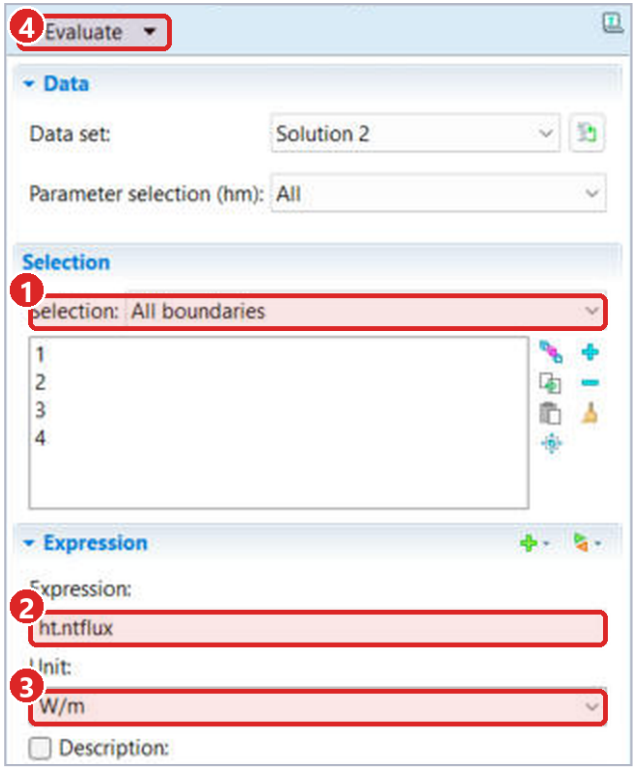
8 One column per point: (0.33,0.33) = T2, (0.67,0.67) = T1, (0.67,0.33) = T3, (0.33,0.67) = T4 (in degC).

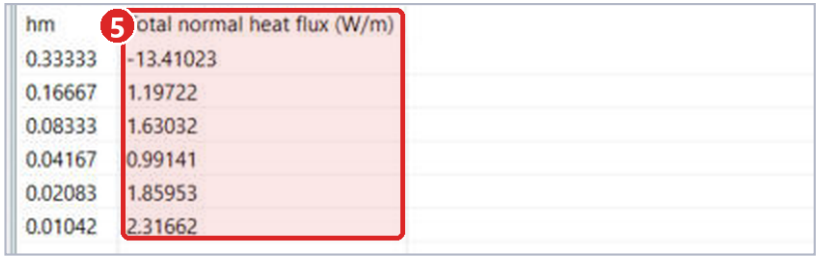
Why?

The values converge to T1 = 490.359, T2 = 359.641, T3 = 438.072 and T4 = 411.928 degC. These agree with the exact series solution of the problem to 5 significant digits, see the results page. The four-unknown finite difference answer of the problem is only an approximation of these values.

Step 15 - Check the global heat balance

Task 5 asks that heat in equals heat out. In COMSOL we add up the normal heat flux over all four edges.





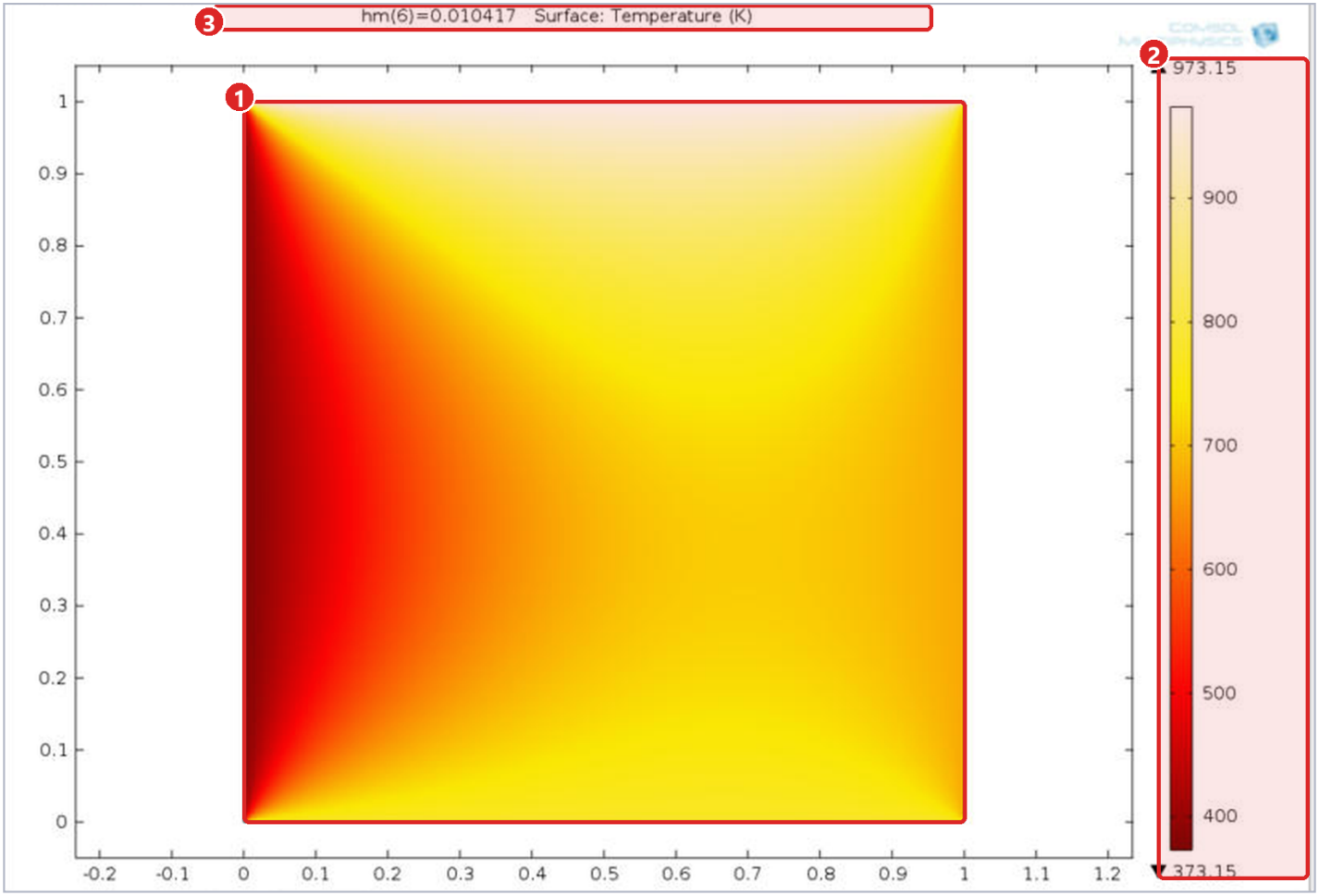
1. Right-click Derived Values > Integration > Line Integration. Data set: Solution 2. Selection: All boundaries (edges 1 to 4).
2. Expression: ht.ntflux, the normal total heat flux through the boundary.
3. Unit: W/m (heat per unit depth of the plate). Click Evaluate.
4. Click Evaluate (the = at the top of the settings). Table 2 opens at the bottom.
5. The sum of the heat flows over all edges, for each mesh (one row per hm).

Why?

If all heat that enters also leaves, this sum is zero. We get -13.4, 1.2, 1.6, 1.0, 1.9, 2.3 W/m, with individual edge flows of several hundred W/m: small compared with the flows, but not zero and not decreasing with mesh refinement. We did not investigate this further; it is a limitation of how COMSOL post-processes boundary fluxes next to the temperature jumps at the corners, so use the temperatures, not the edge fluxes, for this problem. The exact heat balance of the finite difference system is shown on the hand-solution page.

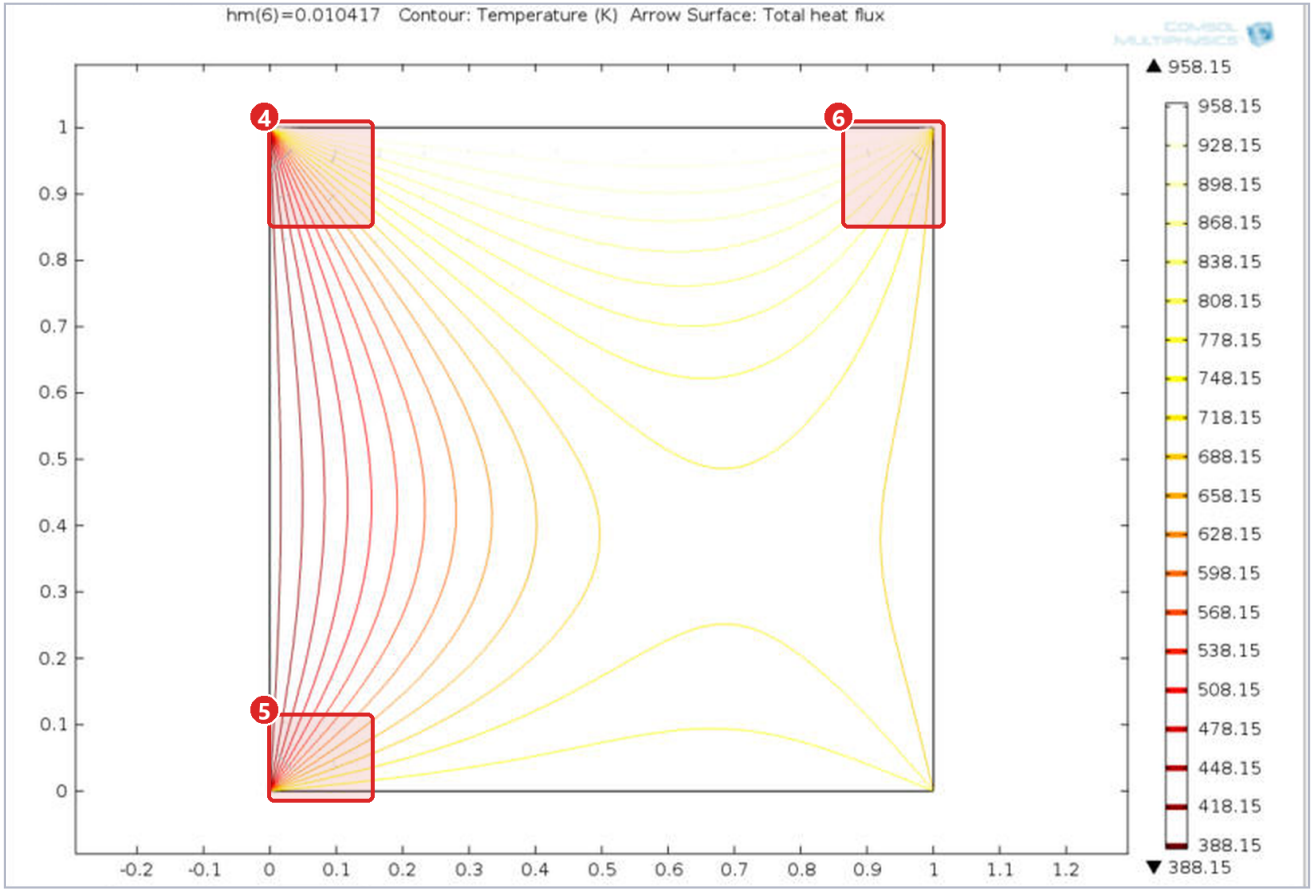
Step 16 - Plot the temperature field and isotherms

Plots help to see whether the result is physically reasonable.



1. Plot group Temperature (ht): surface plot of T. It is hottest at the top (700 C), coldest at the left (100 C).
2. Colour legend in kelvin.
3. Title with the parameter value hm = 1/96.

Step 16 - Plot the temperature field and isotherms



4 Plot group Isothermal contours: lines of equal temperature. They crowd together at the top left corner, where the 700 C and 100 C edges meet.

5 Bottom left corner: 500 C meets 100 C.

6 Top right corner: 700 C meets 400 C.

Why?

The isotherms start perpendicular to the edges... except at the corners, where all isotherms between the two edge temperatures meet in one point. The temperature gradient (and heat flux) is theoretically infinite there. This is why coarse meshes and the finite difference result are least accurate near corners, and why T1 and T2, the interior points that are closest to a corner, show the largest finite difference error.

Results and verification

A. The four temperatures (degC)

Node (x, y)	FDM 3 x 3	Exact series	COMSOL hm=1/3	COMSOL hm=1/96
T1 (2/3, 2/3)	487.5	490.359	490.92	490.359
T2 (1/3, 1/3)	362.5	359.641	358.81	359.641
T3 (2/3, 1/3)	437.5	438.072	438.45	438.072
T4 (1/3, 2/3)	412.5	411.928	411.20	411.928

The exact series solution (superposition of four Fourier series, one per hot edge) is the reference. The 3 x 3 finite difference answer is within 2.9 degC (0.6 % of the temperature range) of it; T1 and T2, which lie closest to the singular corners, have the largest error.

B. COMSOL mesh study (Steps 9 to 14)

hm	DOFs	T1	T2	T3	T4	max |error|
1/3	65	490.923	358.810	438.453	411.196	0.8308
1/6	221	490.384	359.786	438.114	411.742	0.1859
1/12	805	490.358	359.635	438.069	411.898	0.0299
1/24	3045	490.360	359.641	438.072	411.928	0.0010
1/48	11781	490.359	359.641	438.072	411.928	0.0001
1/96	46325	490.359	359.641	438.072	411.928	0.0000

DOFs = degrees of freedom. The 65 for hm = 1/3 follows from the 26-element mesh and 12 boundary edges (1 + 2 x 26 + 12); COMSOL did not print it. COMSOL converges to the exact values: the error is 0.83 degC on the coarsest mesh and 0.001 degC at hm = 1/24.

D. What to remember

- COMSOL (finite elements) and the finite difference method solve the same equation; with a refined mesh both converge to the exact solution.

- The heat balance of the finite difference system is exact (700 in, 700 out). COMSOL's boundary-flux sum is not zero near the corners (Step 15), so judge COMSOL by the temperatures.

- Corner points where two edges at different temperatures meet are singular. The interior values converge nevertheless.

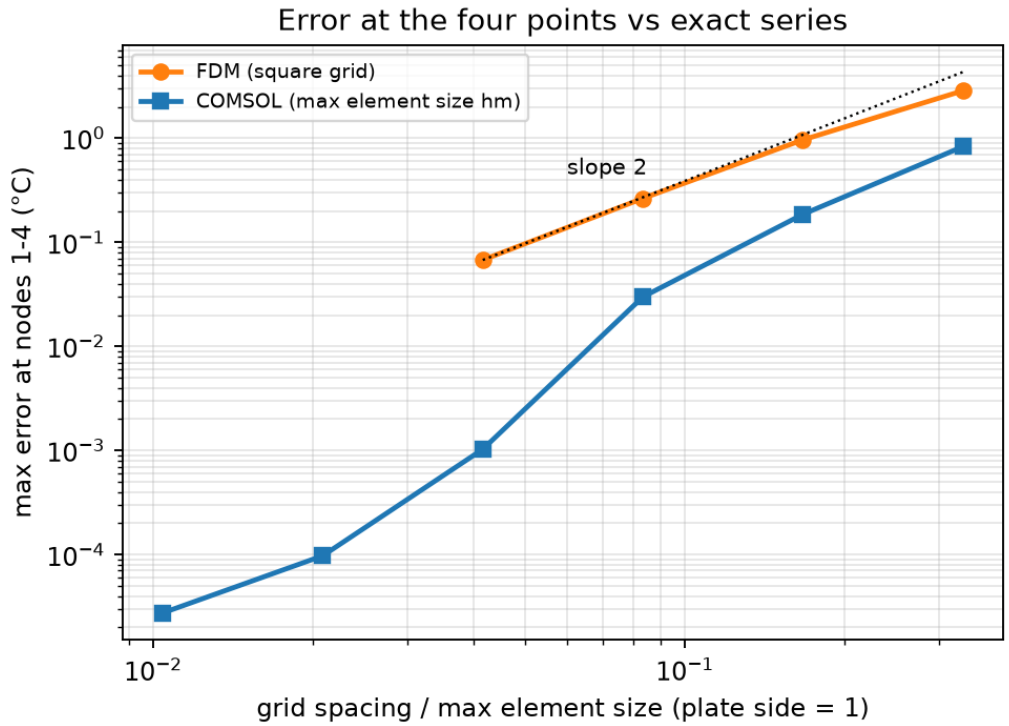
- Quadratic elements make COMSOL converge much faster than the five-point finite difference scheme at the same spacing.

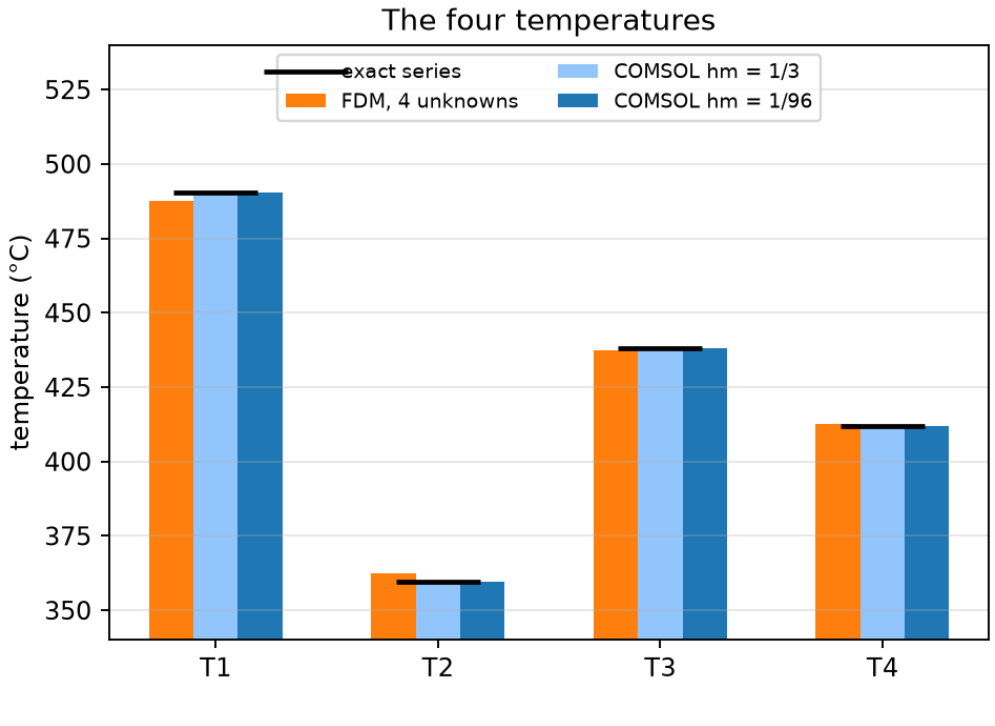
Grid study and convergence

Finite difference grid study (N x N squares)

N	h	T1	T2	T3	T4	max |error|
3	1/3	487.500	362.500	437.500	412.500	2.859
6	1/6	489.394	360.606	437.879	412.121	0.965
12	1/12	490.095	359.905	438.019	411.981	0.264
24	1/24	490.292	359.708	438.058	411.942	0.068

Halving the grid spacing reduces the error about four times (second order). The four-node answer is therefore an approximation good to about 3 degC; with 24 x 24 squares the error is 0.07 degC. At the same spacing COMSOL (quadratic elements) is more accurate than the five-point scheme.





Troubleshooting and tips

The right-click menu does not appear.

Left-click the node once to select it, then right-click again.

A selection list stays empty or contains the wrong edge.

Hover the cursor over the edge until it turns red, click it, then press the + button. If the list shows an edge you did not want, select it in the list and press the minus button. Edges: 1 = left, 2 = bottom, 3 = top, 4 = right.

Typing in a field added to the old text.

Click in the field, press Ctrl+A to select everything, then type the new value.

The temperature is 973 instead of 700.

COMSOL works in kelvin. Write the unit in square brackets: 700[degC]. Without it, 700 means 700 K.

A new row in the sweep table or parameter table does not appear.

For the parametric sweep press the + button below the table and pick hm from the list. For parameters double-click the empty Name cell.

The Point Evaluation table is empty.

Evaluate was not pressed, or the data set is Solution 1 instead of Cut Point 2D 1. Press Evaluate after choosing the data set.

Corner temperatures.

At each corner two edges with different temperatures meet. The problem does not define the corner temperature and the finite difference equations never use it. In COMSOL the Temperature node listed last wins at a corner.

COMSOL asks for a license.

Start COMSOL with the shortcut your installation provides (Start menu, COMSOL 4.3a). On campus the computer must be on the university network or VPN.

Save your work.

File > Save As, choose a folder you can write to. Compare with the supplied file plate_fdm_comsol.mph.